



Appropriating IT outsourcing for IT alignment: An adaptive structuration theory perspective

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ABSTRACT

Previous studies on IT alignment have attempted to address if, and how, the presence of IT outsourcing can help attain IT alignment. In this study, we advance the literature by investigating how organizations appropriate IT outsourcing for achieving IT alignment. Specifically, drawing on the adaptive structuration theory (AST), we construct a model examining factors associated with outsourcing appropriation, including consensus of appropriation (COA) and faithfulness of appropriation (FOA), and how the latter are associated with IT alignment. To test our proposed model, we use a questionnaire survey of 170 IT executives working in Chinese firms. Our results indicate that both shared IT affordance for knowledge acquisition and boundary spanning positively affect COA and FOA. Boundary buffering positively affects the FOA, but not the COA. Further, our results also show the positive effects on IT alignment from COA and FOA, and the significant moderating effects of adoption of cloud computing and the non-significant moderating effects of degree of outsourcing in the relationships. Interestingly, the moderating effects of three cloud computing adoption modes (SaaS, PaaS, and IaaS) on the impact of outsourcing appropriation on IT alignment vary significantly. These findings enrich our understanding towards how to appropriate IT outsourcing for IT alignment and contribute to the literature on IT alignment in IT outsourcing.

1. Introduction

Nowadays, firms capitalize on information technologies (IT) to enhance business value and survive in the volatile global marketplace (Bernat and Karabag, 2019; Queiroz, 2017). The enabling roles of information technology, however, are heavily dependent on IT alignment, i.e., the degree of alignment between organizational IT strategies and business structures (e.g., business mission, objectives, and processes) (Henderson et al., 1996). Prior research has confirmed the strategic value of IT alignment in optimizing IT spending, gaining IT agility, and achieving competitive advantage (Bernat and Karabag, 2019; Reich and Benbasat, 2000; Trang et al., 2021). However, since alignment tends to consume substantial time, human, financial, material, and other assets, the achievement of IT alignment is difficult in reality (Saldanha et al., 2020). To mitigate these restrictions, some firms adopt IT outsourcing solutions that enable firms to gain IT capabilities and resources without increasing new IT spending in deploying hardware and software.

Despite this appealing solution, prior research has revealed some

mixed findings regarding both a positive and negative relationship between IT outsourcing and IT alignment (see Appendix A). Some research has found that outsourcing, by providing leading edge technology for supporting business strategies or by offering talents and new IT solutions to improve IT capabilities, contributes to IT alignment (Park et al., 2017; Wallenburg et al., 2019). However, others have found that outsourcing risks related to improper appropriation of suppliers and service contents may impede IT alignment. For example, ineffective outsourcing activities deriving from improper outsourcing appropriation strategies may introduce unexpected risks, such as opportunistic behavior, added costs, and even business failure (Das and Grover, 2018; Lioliou et al., 2019). As the process of alignment between IT strategies and business structures is dynamic (Sha et al., 2020), unsuitable appropriation may also result in outsourcing inertia or lock-in with unsuitable vendor, which will adversely affect IT alignment (Hirschheim and Sabherwal, 2001; Mol and Kotabe, 2011).

To reconcile these mixed findings, some scholars seek to determine the criteria for suitable outsourcing appropriation (Kunttu, 2017; Liou

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et al., 2011; Perçin, 2019). For example, Perçin (2019) found that outsourcing appropriation was a multi-criteria decision involving cost, delivery, objective, and value. These studies mostly focused on outsourcing appropriation and have not adequately connected outsourcing appropriation to IT alignment. Their findings thus shed few insights on how the effect of outsourcing appropriation on IT alignment may occur. Recently, several attempts have been made to address this problem (Deng et al., 2021). They found that the attitudinal consensus and structural faithfulness towards outsourcing appropriation played an important role in achieving outsourcing objectives (such as IT agility), which is an IT-related issue. Unfortunately, there have been few insights in these studies towards the antecedents of outsourcing appropriation and their impacts on IT alignment.

We construct our research based on the adaptive structuration theory (AST) perspective. In particular, AST provides two important concepts: consensus of appropriation (COA) and faithfulness of appropriation (FOA), as a unique perspective for understanding the appropriation of IT outsourcing (DeSanctis and Poole, 1994; Ko et al., 2021). COA refers to the extent of agreement between IT department and business department regarding the appropriation of IT outsourcing (Deng et al., 2021; Kang et al., 2012). FOA is defined as the extent of which the outsourcing services are appropriated in a way faithful to the structural features (e.g., IT capabilities, technical functions, or service agreement) and spirit (e.g., value and objectives) contained in these services (Deng et al., 2021; Guan et al., 2020).

The two key concepts have important implications for our understanding of how organizations appropriate IT outsourcing for IT alignment. First, given that an important criterion for measuring IT alignment maturity is the partnership between the IT and business departments (Kearns and Sabherwal, 2006; Tallon et al., 2016; Valorinta, 2011), the attitudinal consensus between the two parties towards outsourcing appropriation may be a key factor that affects the partnership quality, which contributes to a better understanding of how organizations appropriate IT outsourcing for IT alignment from the internal relationship perspective. Second, faithful or unfaithful appropriation indicates whether an organization appropriately follows the IT outsourcing supplier's original design intentions (Nguyen et al., 2021), which contributes from the external relationship perspective. Therefore, the COA and FOA perspectives in AST are adaptive for deepening our understanding of how organizations appropriate IT outsourcing for IT alignment.

Taken together, it is essential to explore the antecedents of COA and FOA, and their impacts on IT alignment. In addition, according to AST, the outcomes of structure appropriation may vary depending on the organizational environment (DeSanctis and Poole, 1994; Nguyen et al., 2021). Hence, we also attempt to reveal the moderating factors on the impact of outsourcing appropriation on IT alignment. Specifically, we attempt to answer the following questions:

RQ-1. What factors could promote COA and FOA towards IT outsourcing?

RQ-2. Does COA and FOA improve IT alignment? What are the boundary conditions of these relationships?

Our study makes the following three contributions to the IT outsourcing and AST literature. First, although prior research attempts to connect IT outsourcing to IT alignment, they draw a series of contradictory arguments regarding whether IT outsourcing promotes or hinders IT alignment (Lioliou et al., 2019; Silvius et al., 2013). We seek to address the "IT outsourcing-alignment" paradox from the outsourcing appropriation perspective, and based on AST, divide it into two dimensions, i.e., COA and FOA. Second, although the COA and FOA are likely to be important for IT alignment, it is unclear how organizations form the attitudinal consensus and structural faithfulness towards outsourcing appropriation. Based on AST, we investigate the antecedents of COA and FOA towards outsourcing appropriation from the technology and team characteristics perspectives. In doing so, this study

not only adds to our knowledge on the antecedents of suitable outsourcing appropriation for IT alignment, but also provides valuable suggestions about how organizations achieve the suitable outsourcing appropriation. Third, there has been little focus in previous studies regarding the moderating factors on the impact of outsourcing appropriation on IT alignment. To further deepen our understanding of the impact of outsourcing appropriation on IT alignment, this study also reveals the boundary conditions of these relationships by considering two organizational environment factors: degree of outsourcing and cloud computing adoption, as moderators.

2. Theory and hypothesis development

2.1. Adaptive structuration theory (AST)

AST focuses on studying differences in organizational outcomes that may occur when an advanced technology is used (Kang et al., 2012; Romanow et al., 2018). Based on AST, these differences are more likely to result from the appropriation of social structures embedded in an advanced technology (Kang et al., 2012). AST posits the core concept of appropriation, which is defined as the selection and adaption of social structures contained in an advanced technology (Deng et al., 2021; Ko et al., 2021). It emphasizes that appropriation includes two unique features, i.e., consensus of appropriation (COA) and faithfulness of appropriation (FOA). To be specific, on the one hand, IT users are likely to have different opinions regarding which and how features and functions should be used (Kang et al., 2012). For example, IT departments may mostly focus on some function modules that can integrate with existing modules, while other business departments may pay more attention to some function modules that can align with organizational business structures. On the other hand, IT users can use these functions in a manner faithful or unfaithful to the original intents of designers (Romanow et al., 2018). Together, AST provides a highly relevant lens for studying how organizations appropriate an advanced technology.

Prior research on AST mostly focuses on the IT or IS appropriation, e.g., information and communication technologies (ICT), accounting information system (AIS), and group decision support system (GDSS) (Ko et al., 2021; Nguyen et al., 2021). However, little is devoted to analyzing appropriation of outsourcing from the AST perspective. As posited by Deng et al. (2021), IT outsourcing, as a service mode triggered by advanced technologies, also has the issue of structure appropriation. The structure appropriation towards IT outsourcing involves supplier types (e.g., focused IT service suppliers or IT solution suppliers), contract types (e.g., contract terms, payment schedule agreements, or service-level agreements), delivery types (e.g., on-time delivery or quality delivery), service contents (e.g., system development, operation, or maintenance), and other aspects (Deng et al., 2021; Perçin, 2019). Despite adopting AST to explain how users select and use outsourcing in these studies (Deng et al., 2021), it remains unclear on how users achieve COA and FOA towards IT outsourcing and their impacts on IT alignment.

Different from individual level appropriation towards individual-based technology (Nguyen et al., 2021; Schmitz et al., 2016), we focus on team level appropriation towards outsourcing services triggered by advanced technologies (Kang et al., 2012). As proposed by Kang et al. (2012), IT services adopted by organizations today are more "team-based", rather than "individual-based", because the maximum effect of IT services can be attained if team members use IT services together. Accordingly, we adopt the AST for team-based to examine the antecedents of outsourcing appropriation and its impacts on IT alignment.

Based on AST for team-based, appropriation is mainly affected by two factors, i.e., technology and team characteristics (DeSanctis and Poole, 1994). With respect to the technology factor, we adopt a technology affordance view to understand the role of IT in outsourcing appropriation. Prior research has suggested that technology affordance is a promising lens for predicting technology use or appropriation

(Abhari et al., 2017; Chatterjee et al., 2020; Leonardi, 2013). Technology affordance is defined as the action possibilities and potential that features of a technology offer to its users to attain objectives (Abhari et al., 2017; Lin and Kishore, 2021). Markus and Silver (2008) also argued that affordance could be considered as a driver or enabler of appropriation moves. Given that this study focuses on the team-based appropriation, we adopt the shared affordance perspective (Leonardi, 2013), i.e., the affordance that is shared by team members. Furthermore, drawing on the emerging literature on technology affordance, we focus on shared IT affordance for knowledge acquisition i.e., the knowledge acquisition potential afforded by IT feature use for its users. This is highly relevant for outsourcing appropriation because the knowledge or information owned by teams is the most important determinant for how teams appropriate IT outsourcing (Deng et al., 2021; Kearns and Sabherwal, 2006). As such, we seek to investigate the role of shared IT affordance for knowledge acquisition in outsourcing appropriation.

With respect to the team characteristic factors, we focus on boundary behavior characteristics of IT department. A line of research adopts a boundary view of outsourcing appropriation, and suggests that outsourcing appropriation is a boundary decision where the boundaries of the function and relevant IT activities of IT departments have changed significantly because of the participation of the outsourcing supplier (Du and Pan, 2013). Therefore, we also seek to examine the linkage between boundary behaviors and outsourcing appropriation. Existing research suggests two types of boundary behaviors: boundary spanning and boundary buffering (Zhang et al., 2022). Boundary spanning is conceptualized as activities a team implements to gain resources and support from its external environment (Klueter and Monteiro, 2017). Prior research has widely acknowledged the key roles of team boundary spanning activities, such as forging relationship, knowledge sharing, and lobbying others for support, in obtaining a team access to needed information for decision making (Du and Pan, 2013; Faraj and Yan, 2009). In comparison with boundary spanning, boundary buffering is a disengagement strategy, and refers to behaviors a team undertakes to shield itself against uncertainties and disturbances from the external environment (Wu et al., 2020).

Furthermore, the appropriation process, in essence, reflects a social interaction among social structures, technologies, and users triggered by advanced technologies (Kang et al., 2012). Specifically, based on AST, the adoption of advanced technologies can be considered as the input of a social structure, which coupled with other organizational structures (e.g., business structures), stimulates the social interaction among social structures, technologies, and users, and consequently forms a new social structure that influences organizational outcomes. The process of forming a new social structure via social interaction is termed as structuration (Kang et al., 2012). As proposed by Deng et al. (2021), IT alignment can be regarded as the result of mutual coordination of organizational IT with business structures. However, prior research has largely overlooked IT alignment as an outcome of structuration. Accordingly, we suggest that IT alignment may be the outcome of structuration, and be affected by COA and FOA.

Finally, AST also posits that organizational environment factors play key roles in the structuration process (DeSanctis and Poole, 1994; Nguyen et al., 2021). We expect that the effect of outsourcing appropriation on IT alignment may vary in the two contexts of traditional IT outsourcing and cloud computing. With respect to traditional IT outsourcing, we focus on investigating whether the linkage between outsourcing appropriation and IT alignment is moderated by the degree of outsourcing. With respect to cloud computing, in essence, cloud computing is also a service mode triggered by advanced IT where configurable computing resources or services, e.g., software, hardware, infrastructure, servers, and platform, are stored in a shared pool and are transmitted on demand to clients through network (Gao and Sunyaev, 2019; Mell and Grance, 2010). In contrast with IT outsourcing, cloud computing possesses certain unique features, e.g., pay per use, scalability, elasticity, resource sharing, configurability, and ubiquity, which

contribute to forming low-cost and agile IT solutions (Arvanitis et al., 2017; Mitra et al., 2018). The services provided by cloud computing can be conceptualized as a three-dimensional model, i.e., software as a service (SaaS), platform as a service (PaaS), and infrastructure as a service (IaaS) (Lal and Bharadwaj, 2016). However, earlier research on cloud computing is mostly conducted in a holistic view (Liu et al., 2018). Scant research on this topic, especially for empirical research, have examined the impacts of the three-dimensional model respectively, nor related the three dimensions to IT alignment. As the effectiveness of the mechanism by which outsourcing appropriation impact IT alignment may vary with the contextual variables, research on the impacts of the adoption of cloud computing (including SaaS, PaaS, and IaaS) on the linkage between outsourcing appropriation and IT alignment is necessary to clarify this issue.

2.2. Theoretical model and hypotheses

Drawing on the adaptive structuration theory and relevant literature, we construct a research model to understand the antecedents of COA and FOA, their impacts on IT alignment, and boundary conditions of the impact of outsourcing appropriation on IT alignment (see Fig. 1). Specifically, we examine the antecedents of outsourcing appropriation from the shared IT affordance for knowledge acquisition, boundary spanning, and boundary buffering perspectives. Furthermore, we also investigate the effects of COA and FOA on promoting IT alignment. Finally, we explore the moderating effects of the degree of outsourcing and cloud computing adoption (including SaaS, PaaS, and IaaS) on the impact of outsourcing appropriation on IT alignment.

2.2.1. Shared IT affordance for knowledge acquisition and outsourcing appropriation

Shared IT affordance for knowledge acquisition refers to the knowledge acquisition potential afforded by IT feature use for its users (Abhari et al., 2017; Leonardi, 2013; Nambisan and Baron, 2010). Prior research has confirmed that the knowledge acquisition capability afforded by some IT features, e.g., visibility, reviewability, association, and other aspects, makes it convenient for teams to interact with others (Sun et al., 2019). According to AST, outsourcing appropriation means a social interaction between IT departments, business departments, and suppliers regarding how organizations appropriate outsourcing structures (Deng et al., 2021; Kang et al., 2012; Kearns and Sabherwal, 2006). Some IT tools (e.g., wikis, blogs, or other social networking applications), served as social interaction platforms, can support different forms of knowledge acquisition activities among IT users, such as discussion, knowledge sharing, and social networking, for decision-making processes towards how organizations appropriate IT outsourcing (Lin and Kishore, 2021; Wagner et al., 2014).

From the internal interaction perspective, these knowledge acquisition activities enabled by IT tools help to achieve the joint participation of IT departments and business units in IT and business planning (Kearns and Sabherwal, 2006; Wagner et al., 2014). That is, IT departments can strengthen the understanding of how to better support business strategies through IT, and business departments can deepen the recognition of the intentions of outsourcing strategies made by IT departments. Consequently, both parties are more likely to attain an agreement on how a firm appropriates outsourcing services to achieve business objectives. From the external interaction perspective, these IT tools also provide more communication opportunities for the two sides of IT departments and suppliers (Abhari et al., 2017; Autio et al., 2018), thereby allowing suppliers to clearly understand firms' IT construction demands and reasonably design outsourcing structures for firms' demands. It also enables IT departments to obtain more information on structural features and spirit from outsourcing services, which can help a firm faithfully select suppliers and service contents. Accordingly, we posit that:

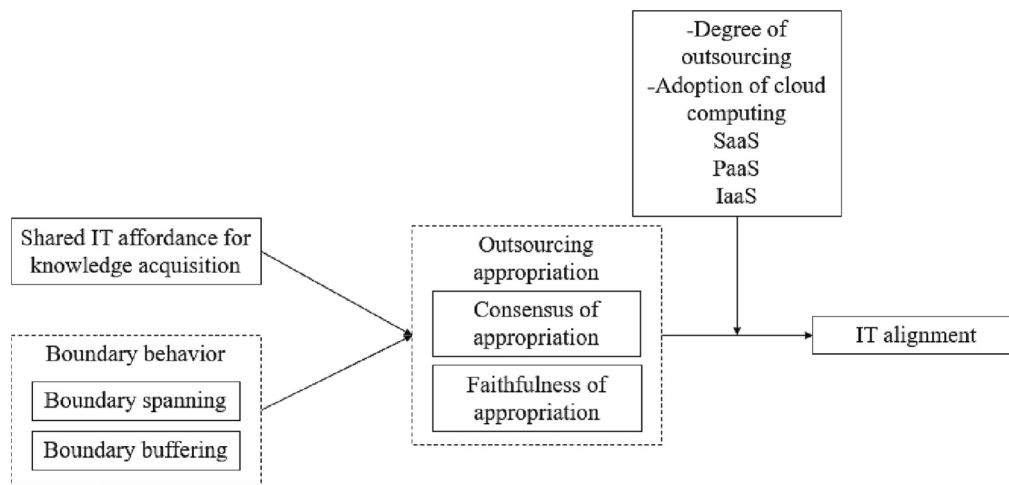


Fig. 1. Research model.

H1a. Shared IT affordance for knowledge acquisition is positively associated with COA.

H1b. Shared IT affordance for knowledge acquisition is positively associated with FOA.

2.2.2. Boundary spanning and outsourcing appropriation

Extant research has adopted the boundary spanning and boundary buffering perspectives to explore how the team boundary management impacts decision-making processes related to IT outsourcing (Valorinta, 2011). A stream of research has widely acknowledged the benefits of cross-boundary interactions (Klueter and Monteiro, 2017), and suggests that some boundary spanning activities, such as forging relationship, knowledge sharing, and lobbying others to obtain support, help teams to gain needed information for outsourcing decision (Hawkins and Reza-zade, 2012; Zhang et al., 2022).

Employees of IT departments engaged in boundary spanning activities are able to acquire the information related to organizational business objectives (Zhang et al., 2022), and based on this information, to design an adaptive IT appropriation planning that can be recognized by business departments. In addition, some cross-interaction behavior of IT departments, e.g., building relationships with business unit or persuading business unit to support the team's work (Klueter and Monteiro, 2017), can strengthen the mutual understanding of each other's needs, and effectively mitigate conflicts arising from the differences of understanding each other's needs, thus contributing to the COA.

Further, boundary spanning activities cover not only the interaction between departments within organization, but also the linkage between firms and suppliers (Deng et al., 2022). Faraj and Yan (2009) proposed that organization department rarely work independently of external support and rarely possessed all of information they require within the organization. As such, firms which employ IT outsourcing businesses tend to adopt some boundary spanning strategies, e.g., setting "task coordinators" (Faraj and Yan, 2009; Zhang et al., 2022). Their work mainly contacts with outsourcing suppliers where they can gain useful information regarding how to use IT outsourcing effectively, which contributes to firms' appropriating IT outsourcing in a faithful way to the original intent of suppliers. Additionally, in the interaction between firms and suppliers, suppliers can understand the outsourcing objectives of firms more comprehensively and develop reasonable outsourcing modes for clients (Teo and Bhattacharjee, 2014), which makes it possible for the client to choose outsourcing services in ways that are intended by suppliers, thereby contributing to the FOA.

H2a. Boundary spanning is positively associated with COA.

H2b. Boundary spanning is positively associated with FOA.

2.2.3. Boundary buffering and outsourcing appropriation

Compared with boundary spanning, which is characterized as an engagement strategy dependent of external support, boundary buffering is a disengagement strategy free from external disturbance and overload needs (Faraj and Yan, 2009; Zhang et al., 2022), which may hinder the COA. As for business units, their strategic focus lies in whether and how business objectives can be achieved, rather than how to achieve objectives through IT outsourcing, mainly because of the complexity and specialization of technological details (Deng et al., 2021). In other words, business units may not always be able to participate in outsourcing appropriation in a positive and proactive manner. If boundary buffering is imposed, business departments are even disengaged from outsourcing appropriation. Their options of outsourcing from business departments are not fully considered and accepted, rendering it difficult to achieve a high level of COA.

In addition, boundary buffering may promote the FOA. In the outsourcing appropriation process, IT departments are frequently affected by disturbances from the external environment, such as outsourcing relationship conflicts or information overload (Schmidt and Uydess, 2012; Zhang et al., 2022), which may bring challenges to the FOA. In particular, an IT department may face lobbying by suppliers and other departments within the client's organization (Lioliou et al., 2019; Teo, 2012). The employees of IT departments often face different or even contradictory outsourcing objectives from senior leaders, business departments, or other departments, which may easily lead to role conflict (Wu et al., 2020). Through buffering, IT departments can protect themselves from uncertainty and turbulence to increase the possibility of rational action. Boundary buffering helps to effectively transfer and manage different or even contradictory demands on IT departments via formal rules or informal norms (Faraj and Yan, 2009; Wu et al., 2020); this is also helpful for IT departments to buffer pressures from the external environment, thereby allowing them to devote more energy to the rational consideration of the service features and service spirit contained in these services from suppliers. Protected by boundary buffering, they can formulate rules to set priorities of the needed service features and service spirit from suppliers via communication and discussion within the department. In sum, boundary buffering is help for the FOA. Thus, we hypothesize that:

H3a. Boundary buffering is negatively associated with COA.

H3b. Boundary buffering is positively associated with FOA.

2.2.4. Outsourcing appropriation and IT alignment

Prior research has suggested that IT alignment can be seen as a newly generated social structure, i.e., the result of the mutual coordination

between organizational IT and business structures (Deng et al., 2021). And based on AST, this new social structure, i.e., IT alignment, may be the outcome of structuration, and be influenced by outsourcing appropriation (including COA and FOA) (DeSanctis and Poole, 1994).

In terms of COA, it means that both the IT and business departments participate in outsourcing appropriation, and reach a consensus regarding how organization appropriate IT outsourcing (Deng et al., 2021; Kang et al., 2012). These processes promote the knowledge integration of IT and business managers, and help them create a “cross function interface”, e.g., brainstorm, conversation, and session discussion (Valorinta, 2011). This offers opportunities for both parties to conduct “collective learning”, which further contributes to providing adaptive IT outsourcing solutions for business strategies. Further, the attitudinal consensus towards IT outsourcing appropriation reflects the partnership between the IT and business departments (Kearns and Sabherwal, 2006; Valorinta, 2011), which enables them to deepen the understanding of each other's values, objectives, and missions. As such, both parties are likely to coordinate their work for work completeness and consistency when there are conflicts in their work, thereby contributing to IT alignment.

In terms of FOA, it reflects the degree to which a client's appropriate IT outsourcing in a way alignment with the general objectives and procedures intended to be provided and affordable by the supplier (Deng et al., 2021; Kang et al., 2012). A high degree of FOA helps clients to build the partnership with suppliers because suppliers believe that they are trustworthy by clients (Teo, 2012). Therefore, it is more likely for suppliers to design suitable IT outsourcing solutions to support clients' business objectives. In addition, prior research has proposed that using outsourcing structures in a similar way enables suppliers to better display their core competencies and advantage resources (Deng et al., 2021), which contributes to achieving IT alignment with the help of IT outsourcing suppliers. Accordingly, we hypothesize:

H4a. COA is positively associated with IT alignment.

H4b. FOA is positively associated with IT alignment.

2.2.5. Moderating effect of the degree of outsourcing

The degree of outsourcing is defined as the degree to which the work associated with IT activities is outsourced (Hanafizadeh and Ahad, 2018). When the degree of outsourcing is high, it is clear that the firm relies more on external IT resources and capabilities to achieve IT alignment. Unreasonable outsourcing appropriation is more likely to lead to the occurrence of negative outsourcing events, such as suppliers' speculative behaviors (Lioliou et al., 2019). Conversely, when the degree of outsourcing is low, internal resources and capabilities make a chief contribution to the construction and maintenance of IT infrastructure, and the impact of outsourcing appropriation on achieving IT alignment is minor. As such, we suggest that the degree of outsourcing may moderate the impact of outsourcing appropriation on IT alignment.

Specifically, in the high degree of outsourcing adoption context, although firms can gain more external IT resources and capabilities, they may need to take more outsourcing risks simultaneously (Lioliou et al., 2019), e.g., the failure of large outsourcing investment and the incompatibility between external services and internal business system. In such case, COA is more crucial for IT alignment. This is because, COA, as the results of a joint agreement between the IT and business departments towards outsourcing appropriation, reflects a prudent outsourcing investment adapted for the technology and business needs of firms (Kang et al., 2012), which contributes to successfully developing and strengthening outsourcing investment capabilities to support IT alignment under the high degree of outsourcing context. Additionally, the high degree of outsourcing may lead to unintentional clusters of fragmented technology silos that constrain the organization from effectively executing routine business activities let alone launching IT alignment (Lu and Ramamurthy, 2011). Prior research has demonstrated that as COA comprehensively considers the technology value and business

value, it could promote perceived usefulness and perceived ease of use for a technology or a service (Kang et al., 2012). Therefore, COA promotes the identification of the value of a technology or a service and avoids making misjudgments on the appropriation of outsourcing services that leads to fragmented silos, which contributes to breaking the fragmented technology silos in the high degree of outsourcing and attains a higher degree of IT alignment.

Furthermore, outsourcing services are designed with the general intention of supporting their clients, such that they can have energies and resources to focus on core competence, rather than on some technical details, e.g., system specifications, implementation, and deployment (Guan et al., 2020; Kroes and Ghosh, 2010). When the degree of outsourcing is high, firms that faithfully adopt these services with such design intention would find that these services dramatically reduce technology development time and translate time into the focus on a higher degree of IT alignment. In addition, the high degree of outsourcing means that outsourcing suppliers may design more adaptive service structures (e.g., technical resources, technical capabilities, or service rules) for their clients and embed their own technical and resource advantages into these structures, because outsourcing suppliers also involve more interests (Deng et al., 2022; Koo et al., 2019; Teo, 2012; Teo and Bhattacharjee, 2014). Under such circumstances, if firms faithfully appropriate these structures, they would promote IT alignment efficiently and effectively with the support of suppliers' advantage resources and capabilities. Accordingly, we hypothesize:

H5a. COA makes a greater contribution to IT alignment when degree of outsourcing is high compared with when it is low.

H5b. FOA makes a greater contribution to IT alignment when degree of outsourcing is high compared with when it is low.

2.2.6. Moderating effect of the adoption of cloud computing

Cloud computing, as an IT service mode, provides technology services such as software, platform, and infrastructure for users through the network (Arvanitis et al., 2017; Gao and Sunyaev, 2019). Given that different types of cloud computing provide some unique characteristics for enterprises, this study attempts to explore the moderating effect of the adoption of cloud computing on the relationship between outsourcing appropriation and IT alignment from the SaaS, PaaS, and IaaS perspectives.

SaaS is a service mode that provides software services for firms through the network in on-demand payments, configurability, and service rental manner (Safari et al., 2015). Although COA enables firms to form an adaptive IT outsourcing solutions for business strategies, the IT solutions jointly determined by the IT and business departments may not necessarily conform to the core competence of suppliers (Benlian, 2009), and may lead to suppliers' speculative behaviors in the process of solution implementation that are not conducive to IT alignment. When firms adopt the SaaS mode, the IT solutions produced by COA can be designed and met with the support of ubiquitous access and faster implementation times, transparent and predictable subscription pricing, and configurable application packages (Benlian, 2009; Liu et al., 2018), thereby contributing to avoiding some speculative behaviors (e.g., transferring businesses to a third party or excessive pricing) and attaining a higher degree of IT alignment.

PaaS is a service mode that provides a software development and deployment platform, including application design, application development, application testing, and application hosting, for users (Cohen et al., 2013). COA can strengthen the interaction between the IT and business departments and form an IT outsourcing solution that reflects business demands needed to be supported by IT. However, the implementation of the IT solution depends on external suppliers and may experience some acknowledged risks, e.g., reliability, data privacy security, and process dependence (Benlian, 2009). PaaS provides firms with the support of the application code base and the tools in a shared manner, and allows them to develop, test, and deploy applications and

services in their own ways (Walraven et al., 2014). Under such circumstances, the IT solutions jointly determined by the IT and business departments enable firms to effectively avoid these unpredicted risks and be met in an expected way, contributing to a higher degree of IT alignment. In addition, PaaS is also characterized by the dynamic extension of the development environment (Walraven et al., 2014). The IT outsourcing solutions formed by COA often cover diversified technical demands and may exceed the upper limits of enterprise IT resource occupancy. Under such circumstances, the role of COA in achieving IT alignment is relatively limited. After the adoption of PaaS, the positive impact of COA on IT alignment may be further strengthened, because PaaS suppliers can monitor the occupancy of service resources, automatically conduct extension based on certain rules, and quickly create the development environment required for tests, demonstration, and up-limit extension in the case of exceeding limit.

IaaS is a service mode in which cloud suppliers provide IT infrastructures, e.g., servers, storage, and computing capabilities, to users via the Internet (Liu et al., 2018; Walraven et al., 2014). The IT solutions formed by the COA depend heavily on the support of the IT infrastructure. With the rapid change of business demands, the IT solutions jointly agreed by the IT and business units need the support of a flexible IT infrastructure. In practice, IaaS suppliers provide some IT infrastructures for firms in a service rental manner (Liu et al., 2018). The service rental model enables firms to acquire hardware resources on demand and achieve a higher degree of resource flexibility (Armbrust et al., 2010), such that firms can reduce the possibility of over-purchase (i.e., waste of resources) and under-purchase (i.e., loss of potential revenues) (Alali and Yeh, 2012). In addition, before the adoption of IaaS, due to the relatively stereotyped or standardized service resources, capabilities, and rules, an IT department for IT alignment may need to frequently interact and coordinate with business units regarding IT outsourcing appropriation, and adjust outsourcing tactics to respond to the changeable business demands (Liu et al., 2018). Therefore, the IT alignment driven by COA is based on the higher adjustment or coordination costs between the IT and business units. After the adoption of IaaS, cloud suppliers deploy these services on their own servers and implement central management through the Internet (Mitra et al., 2018; Nieuwenhuis et al., 2018). When IaaS is adopted, firms are able to quickly obtain hardware resources on-demand through the network and automatically configure resources on suppliers' servers to meet the dynamic needs of outsourcing (Mitra et al., 2018). These technical characteristics reduce the adjustment costs in achieving IT alignment through COA. Therefore, IaaS may impose a significant impact on the relationship between COA and IT alignment by increasing the resource flexibility and reducing the costs. Therefore, we posit that:

H6a. COA makes a greater contribution to IT alignment when SaaS model is adopted compared with when it is not adopted.

H7a. COA makes a greater contribution to IT alignment when PaaS model is adopted compared with when it is not adopted.

H8a. COA makes a greater contribution to IT alignment when IaaS model is adopted compared with when it is not adopted.

Furthermore, based on the above analysis of the characteristics of SaaS, PaaS, and IaaS respectively, we also expect that the faithfulness of the appropriated cloud services to these technical features would achieve IT alignment more conveniently and at lower cost. SaaS is designed with the general characteristics of on-demand payments, configurability, and service rental (Safari et al., 2015), such that firms can conveniently and economically achieve IT alignment. PaaS is designed with the general characteristics of reconfigurability and dynamic extension (Cohen et al., 2013; Walraven et al., 2014), such that firms can attain IT alignment free of path dependence and space up-limit. IaaS is designed with the general characteristics of service rental and flexibility (Liu et al., 2018; Walraven et al., 2014), such that firms can dynamically achieve IT alignment in a flexible and low-cost manner. To realize these

characteristic advantages, cloud suppliers often need to impose certain structures and procedures that their clients can adopt. Hence, firms that faithfully adopt such structures and procedures would greatly enhance the alignment between the appropriated services and business procedures in a flexible, low-cost, convenient, and fast implementation manner, thereby contributing to a higher degree of IT alignment. Accordingly, we hypothesize:

H6b. FOA makes a greater contribution to IT alignment when SaaS model is adopted compared with when it is not adopted.

H7b. FOA makes a greater contribution to IT alignment when PaaS model is adopted compared with when it is not adopted.

H8b. FOA makes a greater contribution to IT alignment when IaaS model is adopted compared with when it is not adopted.

3. Research method

3.1. Data collection

This study adopted a questionnaire-based survey to collect data for testing our proposed research model. We chose the questionnaire method for the following reasons. First, the scales of constructs adopted by our model have been developed by prior research, and their psychometric properties have also been well confirmed. Second, a questionnaire survey enabled us to obtain enough samples from large quantities of respondents, which contributed to a more robust and generalizable conclusion. Third, considering the nature of questionnaire, e.g., anonymous and voluntary, social desirability bias can be effectively mitigated.

To ensure its validity and reliability, our questionnaire was developed using a four-step process. First, the questionnaire items were adapted from existing literature in which psychometric properties have been established. Second, we conducted backward translation to ensure the agreement regarding the meaning of the items between the Chinese version and the English version of the questionnaire. We also invited four experts on IT outsourcing to conduct the pre-test for content validity. Further, we also organized semi-structured interviews with one professor and three doctoral students as interviewers and five IT managers as interviewees, aiming to help us check the words, grammar, and clarity in these items. Third, given that our discussion on COA and FOA focused on IT outsourcing appropriation situation, rather than traditional IT or IS appropriation situation in extant studies, there may exist the issues of content overload and ambiguity in our scales. For this reason, we adopted item-sorting, and consequently found no significant misclassification. Fourth, we implemented a pilot test with 32 IT managers or above who have ever been in charge of or participated in IT outsourcing business, and removed some low item-to-total correlation items. By evaluation and revision, the validity and reliability of our questionnaire have been effectively improved. Finally, we further examined the validity and reliability of our scales using some analytical tools. The specific testing results are shown in Data analysis and results section.

Our targeted respondents are mostly from IT departments in some manufacturing and service firms. Based on the principle of theoretical sampling, IT department managers or other unit executives that took charge of or participated in outsourcing businesses were more suitable as our targeted respondents, because they were the key participants and boundary spanners for IT alignment through outsourcing appropriation. For this reason, we set a screening item at the beginning of our questionnaire, i.e., "If your position belongs to the IT department team leader, IT department manager, IT department management, or other unit executives that take charge of or participated in outsourcing businesses, please continue to answer the following questions, otherwise the questionnaire ends".

The questionnaire was mainly sent through two channels. First, we

set an online survey supported by a top survey provider in China. The service afforded by this provider included online survey, management consulting, customization solutions, big data analytics and listening, and other aspects. At present, their services have covered over 10,000 firms, institutions, universities, governments, and other subjects. Especially, they have owned a China's leading database of user profiles by cooperating with some large-scale Internet enterprises, e.g., Google, Baidu, Alipay, and Tencent, and thus largely cover Chinese netizens. Based on the alignment principle of our research context with user panel in the database of this provider, we determined our targeted informants, and sent survey invitations to them. Given that we conducted eight rounds of follow-up reminders, there may exist the issue of non-response bias. Therefore, we adopted *t*-tests to compare the work years, industries, firm sizes, and numbers of employee of the first quarter and last quarter respondents. Consequently, we found no non-response bias between the two quarter respondents at the 0.05 level in all the four *t*-tests.

Second, following the snowball sampling method, we sent our survey invitations to respondents via team work and personal relationships in the IS field. For instance, we requested well known outsourcing and cloud service providers (e.g., Alibaba) to help us distribute questionnaires. Given that our data was collected from the two different channels, we executed an independent-samples *t*-test. Our results indicated no significant difference on key sample features. Hence, we combined the two sample groups as a single sample.

To assess whether our respondents answered these items carefully and accurately, we set some rules and criteria. First, respondents were required to take charge of or participate in outsourcing or cloud computing businesses in their work. In the data processing process, we dropped unqualified samples with no IT outsourcing or cloud computing business experience. Second, we included some screening items in our questionnaire to confirm whether our questionnaire was responded by respondents carefully. For example, respondents selected no adoption of cloud computing in the first option, while they also selected amounts of cloud computing budget in the second option. Obviously, this is a contradictory answer. These samples that filled in the questionnaire in a logical contradiction way were excluded.

Through the above processes, we finally obtained 170 valid responses. In these valid responses, respondents that were from the manufacturing sector accounted for 42.35 %, and others that were from the services sector accounted for 57.65 %. Respondents that had more than three years of working experience accounted for 75.88 %. Their positions were mostly team leader of IT department or above (approximately accounting for 98.24 %). These sample characteristics ensured that our selected respondents were familiar with or in charge of their outsourcing or cloud computing businesses. Therefore, our samples were representative for our research question. The detailed demographics of the firms were depicted in Table 1.

3.2. Measures

All measurement items, including shared IT affordance for knowledge acquisition, boundary spanning, boundary buffering, COA, FOA, and IT alignment, have been developed by existing literature in which psychometric properties were well confirmed. We further adapted certain items aiming to promote the alignment of these items and our research focus. In addition, we adopted a 7-point response format to measure all items.

The shared IT affordance for knowledge acquisition scale was derived from Abhari et al. (2017) and Leonardi (2013). It consisted of six items covering the knowledge acquisition potential afforded by IT tools used by IT innovation collaborators in two aspects, i.e., knowledge searching and knowledge integration. An example for knowledge searching was "When our innovation collaborators (such as business departments, vendors, and customers) and our IT department seek IT innovation activities, the IT used enables us to: easily access data between ours and innovation collaborators' systems." An example for

Table 1

Demographics of the firms (number of samples = 170).

Industry		Working experience (years)	
Manufacturing Service	42.35 %	<1	7.06 %
	57.65 %	1–3	17.06 %
		3–5	33.53 %
		5–10	29.41 %
		>10	12.94 %
Firm size		Number of IT staff	
<100 100–500 501–1000 >1000	19.41 %	<3	4.71 %
	37.65 %	3–5	11.18 %
	20.00 %	6–10	18.82 %
	22.94 %	11–20	21.18 %
		21–50	20.59 %
		>50	23.53 %
The proportion of work associated with system development that is outsourced		The proportion of work associated with system and infrastructure maintenance and support that is outsourced	
<20 % 20 %–40 % 41 %–60 % 61 %–80 % >80 %	18.24 %	<20 %	21.76 %
	39.41 %	20 %–40 %	35.88 %
	26.47 %	41 %–60 %	27.06 %
	11.76 %	61 %–80 %	11.76 %
	4.11 %	>80 %	3.53 %
Whether adopt IT outsourcing		The proportion of cloud computing spending to IT budget	
Yes No	100.00 %	<10 %	12.94 %
	0.00 %	10 %–20 %	37.06 %
		21 %–30 %	30.00 %
		31 %–50 %	12.94 %
		>50 %	7.06 %

knowledge integration was "When our innovation collaborators (such as business departments, vendors, and customers) and our IT department seek IT innovation activities, the IT used enables us to: easily aggregate relevant information." The coefficient alpha of this scale was 0.866.

The boundary spanning and boundary buffering scales were measured using four items adopted by Faraj and Yan (2009), and were mainly concerned with the boundary spanning and boundary buffering behavior of IT departments with the IT department as the boundary. An example for boundary spanning was "To what extent does the IT department encourage its staff to solicit information and resources from elsewhere in and/or beyond the department". An example for boundary buffering was "To what extent are outside pressures deflected or absorbed so that the IT department can work free of interference". The coefficient alpha of the two scales was 0.760 and 0.732 respectively.

The COA scale, which consisted of five items measuring the agreement level between IT department and business unit towards appropriation of IT outsourcing, was derived from Deng et al. (2021). The FOA scale, which involved six items measuring the structural faithfulness in two aspects: structural features (e.g., IT resources and capabilities, technical functions, or service rules) and spirit (e.g., service philosophy, value, and objectives), was derived from Deng et al. (2021). An example for COA was "Between our company's IT department and business unit: there is mutual understanding in our company on how we should adopt IT outsourcing strategy". An example for FOA was "As for the supplier and the IT outsourcing services we choose from them: our IT outsourcing strategy objectives are matched with the vendor's beliefs or strategic objectives". The coefficient alpha of the two scales was 0.881 and 0.891 respectively.

The IT alignment scale, which consisted four items measuring the degree to which the alignment of IT strategies with business strategies,

was derived from Preston and Karahanna (2009). An example for IT alignment was “Overall, the IS in our organization meets the needs of the business strategy”. The coefficient alpha of this scale was 0.872.

Regarding the measurement of the degree of outsourcing, we used the question “How much of the work related to the system development (system specifications, implementation, and deployment) is outsourced?” with options of “less than 20%” as “1”, “20 %–40 %” as “2”, “41 %–60 %” as “3”, “61 %–80 %” as “4”, and “more than 80%” as “5”. Regarding the measurement of the adoption of cloud computing, we used the question “Does your company use cloud computing services in the following areas (multiple options)?” with options of “software” as “1”, “platform” as “2”, “infrastructure” as “3”, and “none of the above” as “4”.

Furthermore, by reviewing literature on IT outsourcing and IT alignment, we considered types of industry, numbers of employee, numbers of IT staff, position, working years as control variables, because these factors have been shown to potentially affect IT alignment (Coltman et al., 2015).

4. Results analysis

4.1. Validity and reliability

To assess the reliability and validity of our questionnaires, this study used Lisrel 8.8 as a statistical analysis tool to conduct confirmatory factor analysis. The expected six-factor model including the shared IT affordance for knowledge acquisition, boundary spanning, boundary buffering, COA, FOA, and IT alignment displayed an acceptable fit (RMSEA = 0.067, $\chi^2 = 638.27$, $df = 362$, $\chi^2/df = 1.76$, CFI = 0.97, NNFI = 0.97, IFI = 0.97). The χ^2/df value was <2. The CFI, NNFI, and IFI values were >0.90 respectively, while the RMSEA value was 0.067, lower than the threshold value of 0.08. Therefore, construct validity of our model was basically acceptable.

We conducted construct reliability tests in two ways. First, we used the indicator of Cronbach's alpha (α) to assess internal consistency reliability. Second, we also employed the indicator of composite reliability (CR) to evaluate internal consistency reliability. The results of the

Table 2
Results of reliability and convergent validity.

Constructs	Items	Loadings	α	CR
Shared IT affordance for knowledge acquisition (SA)	SA01	0.67	0.866	0.869
	SA02	0.65		
	SA03	0.76		
	SA04	0.80		
	SA05	0.70		
	SA06	0.76		
Boundary spanning (BS)	BS01	0.68	0.760	0.767
	BS02	0.77		
	BS03	0.62		
	BS04	0.61		
Boundary buffering (BB)	BB01	0.60	0.732	0.739
	BB02	0.73		
	BB03	0.59		
	BB04	0.65		
Consensus of appropriation (COA)	COA01	0.76	0.881	0.882
	COA02	0.78		
	COA03	0.73		
	COA04	0.82		
	COA05	0.78		
Faithfulness of appropriation (FOA)	FOA01	0.74	0.891	0.892
	FOA02	0.78		
	FOA03	0.72		
	FOA04	0.78		
	FOA05	0.76		
	FOA06	0.78		
IT alignment (ALI)	ALI01	0.81	0.872	0.873
	ALI02	0.80		
	ALI03	0.82		
	ALI04	0.75		

two indicators for each construct are listed in Table 2. Results showed that all values of α and CR were >0.73, indicating adequate reliability.

To verify construct convergent validity, we regarded the value of item factor loading as a judgement criterion. The results of factor loading values are also shown in Table 2. Most loading values were above 0.70, and only few items had relatively lower values ranging from 0.59 to 0.68 (but over the cutoff point of 0.50). Therefore, convergent validity of all constructs was acceptable.

Further, we conducted discriminant validity tests as follows. First, we calculated the values of average variance extracted (AVE) for all constructs to evaluate discriminant validity. The results of descriptive statistics for each construct, including mean, standard deviation, correlations, and AVE, are presented in Table 3. Results showed that the square roots of AVE for all constructs were higher than the correlations with other constructs, showing acceptable discriminant validity. Second, we examined discriminant validity by using correlation indicators. As shown in Table 3, all correlations between the construct and others were lower than 0.7, thus indicating good discriminant validity. Third, we employed chi-square difference tests to evaluate discriminant validity. Specifically, we compared one CFA model freeing correlations between the two constructs to another CFA model fixing a perfect correlation. According to six constructs adopted by our model, we conducted 15 round comparisons between the two constructs. Through computing, we adopted the Bonferroni correction p-value of 0.033333 because we conducted 15 rounds of repeated test (Kroes and Ghosh, 2010). Results showed that each round chi-square difference comparison was significant, therefore showing acceptable discriminant validity.

4.2. Common method bias

Our study experienced a threat of common method bias, because we adopted self-reported data collected by subjective instruments. For this reason, to check for common method bias, we subsequently employed four techniques that have been widely used in prior research. First, as suggested by Podsakoff et al. (2003), we adopted some ex-ante procedural remedy measures, e.g., disrupting the item order, ensuring anonymity, and setting screening question. Second, we conducted Harman's single factor test, aiming to assess whether common method bias existed or not. The results indicated that the variance explained by the first factor accounted for 38.3 % of total variances, which was far <50 %. As such, the single factor that may explain the majority of total variances does not exist. Third, we use the indicators of variance inflation factor (VIF) to check for the collinearity issue. The VIF values were all lower than the threshold value of 3.0, thereby showing that collinearity was not a serious problem. Fourth, we constructed a single-factor analysis model, including shared IT affordance for knowledge acquisition, boundary spanning, boundary buffering, COA, FOA, and IT alignment, and conducted a confirmatory factor analysis. The constructed single-factor model displayed an unsatisfactory fit (RMSEA = 0.141, CFI = 0.86, NNFI = 0.85, IFI = 0.86). Conversely, the expected six factor model attained a significant improvement consistent with our data. In sum, the threat of common method bias in this study was not a serious problem.

4.3. Hypotheses testing

Hypotheses testing was conducted using hierarchical regression analysis. The regression results of the antecedents of outsourcing appropriation are shown in Table 4. As shown in Table 4, M1-2 indicated that shared IT affordance for knowledge acquisition ($\beta = 0.266$, $p < 0.01$) and boundary spanning ($\beta = 0.202$, $p < 0.05$) were positively associated with COA. Thus, H1a and H2a were supported. However, the impact of boundary buffering on the COA was not significant. Thus, H3a was not supported. In M2-2 of Table 4, shared IT affordance for knowledge acquisition ($\beta = 0.289$, $p < 0.01$), boundary spanning ($\beta = 0.161$, $p < 0.05$), and boundary buffering ($\beta = 0.252$, $p < 0.01$) were

Table 3
Descriptive statistics, correlations and AVE.

Variables	1	2	3	4	5	6	7	8	9	10
1.SA	0.725									
2.BS	0.506**	0.673								
3.BB	0.555**	0.589**	0.645							
4.COA	0.448**	0.429**	0.395**	0.775						
5.FOA	0.516**	0.466**	0.497**	0.612**	0.760					
6.ALI	0.467**	0.437**	0.393**	0.574**	0.688**	0.795				
7.DE	0.096	0.004	0.004	0.140	0.113	0.126	NA			
8.SaaS	0.171*	0.096	0.170*	0.078	0.003	−0.023	0.017	NA		
9.PaaS	0.216**	0.154*	0.177*	0.199**	0.153*	0.185*	0.163*	0.027	NA	
10.IaaS	0.226**	0.257**	0.186*	0.188*	0.226**	0.281**	−0.024	−0.004	0.060	NA
Mean	5.178	5.057	5.019	5.202	5.203	5.269	2.418	0.530	0.690	0.540
SD	0.959	0.935	0.939	0.996	0.962	1.000	0.988	0.501	0.465	0.500

Note: ** $p < 0.01$; * $p < 0.05$; DE means degree of outsourcing; bold numbers mean the square roots of the AVE.

Table 4
The regression results of the antecedents of outsourcing appropriation.

	COA		FOA	
	M1-1	M1-2	M2-1	M2-2
Control variables				
Position	−0.021 (0.056)	−0.013 (0.049)	0.018 (0.053)	0.029 (0.043)
Industry	0.207 (0.154)	0.181 (0.135)	0.347* (0.147)	0.317** (0.119)
Numbers of employee	−0.108 (0.075)	−0.039 (0.067)	−0.044 (0.072)	0.026 (0.059)
Numbers of IT staff	0.136* (0.061)	0.065 (0.055)	0.134* (0.058)	0.058 (0.048)
Working years	0.145† (0.075)	0.133* (0.066)	0.111 (0.071)	0.108† (0.059)
Antecedents				
SA		0.266** (0.086)		0.289** (0.076)
BS		0.202* (0.092)		0.161* (0.081)
BB		0.149 (0.095)		0.252** (0.084)
Adjusted R ²	3.6 %	26.2 %	5.9 %	38.4 %
ΔR ²		23.3 %**		32.6 %**
F statistics	2.245	8.503	3.138	14.171

Note: ** $p < 0.01$; * $p < 0.05$; † $p < 0.10$; S.D. values were included in parentheses.

positively associated with FOA. Thus, [H1b](#), [H2b](#), and [H3b](#) were supported.

[Table 5](#) showed the regression results of the effects of outsourcing appropriation and moderating effects of the degree of outsourcing and SaaS, PaaS, and IaaS. M3-2 indicated a positive impact on IT alignment from COA ($\beta = 0.242$, $p < 0.01$) and FOA ($\beta = 0.541$, $p < 0.01$) respectively. As such, [H4a](#) and [H4b](#) were supported.

With respect to the moderating effects of the degree of outsourcing and SaaS, PaaS, and IaaS, we examined the significance of interaction term between independent variables and moderators to judge whether the moderating effects is significant or not. Following the advice given by [Cohen et al. \(2013\)](#), we also plotted the moderating effects in [Figs. 2 to 5](#), and analyzed them using simple slope analysis.

H5 hypothesized that outsourcing appropriation make a greater contribution to IT alignment when degree of outsourcing is high. M4-2 indicated a non-significant impact on IT alignment from the interaction term between COA and degree of outsourcing ($\beta = -0.034$, $p > 0.10$) and the interaction term between FOA and degree of outsourcing ($\beta = -0.022$, $p > 0.10$) respectively. Thus, [H5a](#) and [H5b](#) were not supported.

H6, H7, and H8 hypothesized that outsourcing appropriation make a greater contribution to IT alignment when SaaS, PaaS, and IaaS model are adopted compared with when they are not adopted. Specifically, M4-

Table 5
The effects of outsourcing appropriation and moderating effects of the degree of outsourcing and SaaS, PaaS, and IaaS.

	IT alignment			
	M3-1	M3-2	M4-1	M4-2
Control variables				
Position	−0.006 (0.055)	−0.011 (0.040)	0.004 (0.041)	0.017 (0.042)
Industry	0.392* (0.154)	0.154 (0.114)	0.151 (0.113)	0.124 (0.116)
Numbers of employee	−0.046 (0.075)	0.004 (0.055)	0.029 (0.056)	−0.005 (0.057)
Numbers of IT staff	0.120* (0.061)	0.015 (0.045)	−0.015 (0.048)	0.008 (0.049)
Working years	0.110 (0.075)	0.015 (0.055)	0.008 (0.056)	0.020 (0.057)
Independent variables				
COA		0.242** (0.070)	0.232** (0.071)	0.966** (0.340)
FOA		0.541** (0.074)	0.512** (0.074)	−0.124 (0.340)
Moderating variables				
DE			0.031 (0.057)	0.120 (0.108)
SaaS			−0.085 (0.110)	0.157 (0.201)
PaaS			0.121 (0.125)	0.036 (0.206)
IaaS			0.253* (0.117)	0.101 (0.206)
Interactions				
DE × COA				−0.034 (0.106)
SaaS × COA				−0.323† (0.167)
PaaS × COA				−0.512* (0.209)
IaaS × COA				−0.212 (0.162)
DE × FOA				−0.022 (0.100)
SaaS × FOA				0.172 (0.175)
PaaS × FOA				0.568** (0.211)
IaaS × FOA				0.290† (0.168)
Adjusted R ²	5.0 %	49.6 %	50.4 %	51.7 %
ΔR ²		43.9 %**	45.8 %**	3.5 %
F statistics	2.762	24.733	16.584	10.509

Note: ** $p < 0.01$; * $p < 0.05$; † $p < 0.10$.

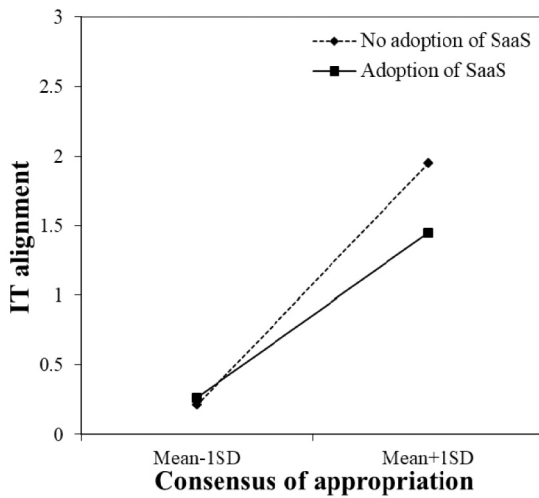


Fig. 2. COA-SaaS-IT alignment.

2 showed a significant and negative impact on IT alignment from the interaction term between COA and SaaS ($\beta = -0.323$, $p < 0.10$) and the interaction term between COA and PaaS ($\beta = -0.512$, $p < 0.05$) respectively. As depicted in Figs. 2 and 3, the positive effect of COA on IT alignment is weaker when SaaS (and PaaS) is adopted than that it is not adopted. These results were opposite to H6a and H7a. Further, the coefficient of the interaction term between COA and IaaS was not significant. Thus, H8a was not supported.

As for the FOA, M4-2 showed a non-significant impact on IT alignment from the interaction term between FOA and SaaS ($\beta = 0.172$, $p > 0.10$). Thus, H6b was not supported. In addition, M4-2 also showed a positive impact on IT alignment from the interaction term between FOA and PaaS ($\beta = 0.568$, $p < 0.01$) and the interaction term between FOA and IaaS ($\beta = 0.290$, $p < 0.10$) respectively. As depicted in Figs. 4 and 5, the positive impact of FOA on IT alignment is stronger when PaaS (and IaaS) is adopted than that it is not adopted. Thus, H7b and H8b were supported.

5. Discussion

5.1. Theoretical contributions

IT outsourcing, as an important external source of firms' IT resources and capabilities, has long been regarded as a promising way to attain IT alignment. However, firms sometimes hesitate to seek IT outsourcing

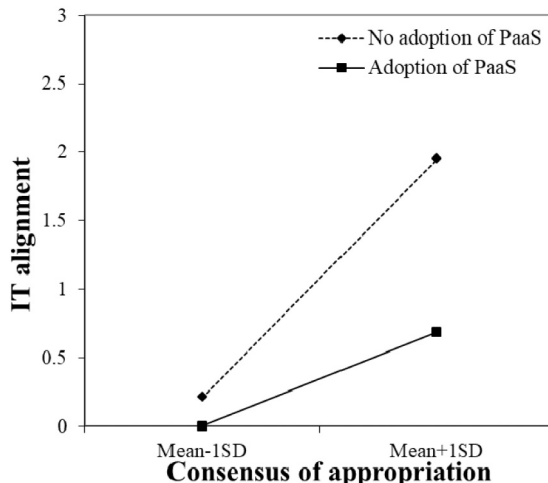


Fig. 3. COA-PaaS-IT alignment.

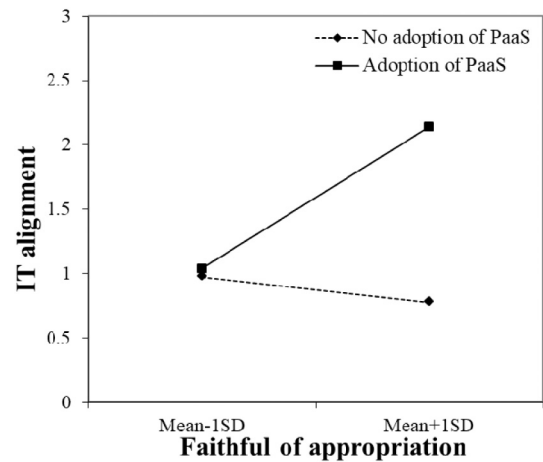


Fig. 4. FOA-PaaS-IT alignment.

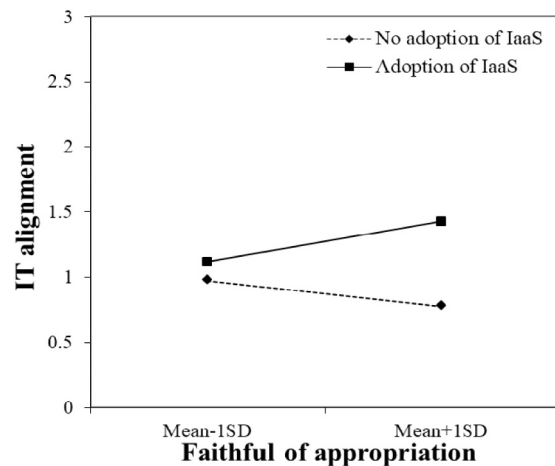


Fig. 5. FOA-IaaS-IT alignment.

solutions to attain IT alignment because of outsourcing risks arising from improper outsourcing appropriation (Deng et al., 2021; Lioliou et al., 2019). Therefore, researchers turn to the topic on the criteria for suitable outsourcing appropriation. This study empirically extends extant theories on IT outsourcing and AST by investigating the antecedents of outsourcing appropriation (referring to COA and FOA in this study), their impacts on IT alignment, and boundary conditions of the impacts of outsourcing appropriation on IT alignment.

Our study contributes to the IT outsourcing and AST literature in the following several ways. First, different from prior literature that attached great importance to the outcomes of outsourcing appropriation (Deng et al., 2021), based on AST, our study revealed the antecedents of outsourcing appropriation from the technology and team characteristic perspectives. From the technology perspective, our results suggest that shared IT affordance for knowledge acquisition promotes both COA and FOA. Corresponding with previous literature that focused on the positive roles of IT tools in decision-making processes (Sun et al., 2019), we find that the knowledge acquisition potential afforded by the IT contributes to achieving the attitudinal consensus and structural faithfulness towards outsourcing appropriation decisions. From the team characteristic perspective, our results also suggest that boundary spanning has a positive effect on both COA and FOA. Furthermore, boundary buffering has a positive effect on FOA, but no significant effect on COA. One possible explanation for the non-significant result is that boundary buffering may impose both positive and negative impacts on COA. On the one hand, boundary buffering may indeed cause conflicts and

discourage involvement from other departments within the organization, thus damaging the COA (Wu et al., 2020). However, on the other hand, boundary buffering, especially to buffering behavior relating to external suppliers, may protect IT departments against disturbances from suppliers' lobbying, thereby improving the behavioral rationality and efficiency of IT departments for the achievement of COA (Faraj and Yan, 2009; Zhang et al., 2022). Overall, this finding reveals that boundary behaviors, including boundary spanning and buffering, differentially change the COA and FOA.

Second, compared with the contradictory arguments on the linkage between IT outsourcing and alignment in prior literature (Silvius et al., 2013), this study provided in-depth insights for addressing the "IT outsourcing-alignment" paradox from the COA and FOA perspectives. Our empirical results indicate that COA and FOA contribute to promoting IT alignment respectively. These findings show the importance of the attitudinal consensus and structural faithfulness towards IT outsourcing appropriation on achieving IT alignment, thus mitigating the concern for outsourcing risks arising from improper outsourcing appropriation in existing literature. In addition, although extant studies have highlighted the key roles of COA and FOA in achieving outsourcing objectives (e.g., IT agility) (Deng et al., 2021), few has related the attitudinal consensus and structural faithfulness towards IT outsourcing to IT alignment. This study empirically reveals that the IT alignment can be equally viewed as a key outcome of COA and FOA.

Third, this study further reveals the boundary conditions of the effects of outsourcing appropriation on IT alignment. Scant research has attempted to investigate the impacts of COA and FOA on IT alignment, nor has focused on the boundary conditions of these relationships. This study seeks to fill these gaps by investigating the moderating roles of degree of outsourcing and cloud computing adoption. Our results find that the moderating effects of the degree of outsourcing on the effect of outsourcing appropriation on IT alignment are not significant. This finding further supports the previous point that the effects of outsourcing appropriation on IT alignment are quite significant and close. In other words, regardless of the high or low degree of outsourcing, based on the consideration of risk aversion, firm executives in both cases attach great importance to the attitudinal consensus and structural faithfulness towards IT outsourcing to avoid the outsourcing risks arising from unreasonable outsourcing appropriation, and further attain IT alignment by leveraging the key roles of COA and FOA.

Further, differing from prior research on the adoption of cloud computing from a holistic view (Liu et al., 2018), our study also empirically investigates the moderating effects of the different dimensions of cloud computing adoption on the effects of outsourcing appropriation on IT alignment. Our results indicate that when SaaS or PaaS is adopted, the impact of the COA on IT alignment is lower than that when they are not adopted. One possible explanation for this is that SaaS and PaaS models are more standardized (Fardinpour et al., 2020; Singh et al., 2016), and it is therefore harder for them to achieve on-demand personalization. Thus, although COA is reached, the failure of SaaS or PaaS in the configuration based on the individual needs of clients might lead to the relatively limited improvement of IT alignment. However, if SaaS or PaaS is not adopted and more traditional outsourcing or IS development by oneself are considered instead, it might be easier for the outsourcing supplier or internal IT department to meet individual needs, and it may be easier for the COA to achieve IT alignment. Comparably, we also find a non-significant impact on IT alignment from the interaction term between the COA and IaaS. In sum, these results suggest that when cloud computing is adopted at a higher application level rather than at an infrastructure level, the positive role of the COA in alignment may be weakened. Our results also suggest that when PaaS or IaaS is adopted, the positive impact of the FOA on IT alignment is higher than that when they are not adopted, but the moderating role of SaaS is not significant. The non-significance associated with SaaS suggests that the role of the FOA in IT alignment does not vary significantly with the adoption of SaaS. On the one hand, the

adoption of SaaS may indeed strengthen the positive effect of the FOA on IT alignment owing to its quick and low-cost configurability and extendable technical advantages (Khayer et al., 2020). On the other hand, due to its relative standardization and difficulties in adapting to personalized business needs (Fardinpour et al., 2020), the adoption of SaaS may also weaken the positive role of the FOA on IT alignment. It may therefore be the combination of positive and negative effects that leads to the non-significance of the moderating effects of SaaS. In sum, the overall results indicate that when cloud computing is adopted at the high infrastructure or platform levels rather than at the application level, the FOA is more conducive to achieving IT alignment. Therefore, our study provides in-depth insights regarding how different dimensions of cloud computing adoption impose a significant impact on this relationship.

5.2. Managerial implications

Our study offers some valuable practical guidance for department managers of outsourcing users at various levels, e.g., IT department managers and business unit managers, and outsourcing suppliers. First, our results enable organizations to deepen the understanding of what factors promote the attitudinal consensus and structural faithfulness towards IT outsourcing appropriation. Specifically, our results highlight the importance of the knowledge acquisition potential afforded by IT tools on the consensus and faithful appropriation towards IT outsourcing. Therefore, firm executives should attach great importance to the roles of IT support in outsourcing appropriation decisions, and especially, regard some IT functional modules aiming to improve organizations' knowledge acquisition capabilities as a key criterion of IT investments. Our results also show that the boundary behaviors for teams play crucial roles in outsourcing appropriation. On one hand, the IT department should conduct some boundary spanning behaviors together with other departments and suppliers, such as forging relationship, knowledge sharing, and lobbying others to obtain support, to strengthen the mutual understanding of how outsourcing services are appropriated. On the other hand, the IT department should be sensitive to the risks with boundary work to protect itself against disturbances from suppliers' lobbying in the outsourcing appropriation process.

Second, our results emphasize the crucial role of outsourcing appropriation, and the key point for achieving IT alignment via IT outsourcing is to promote COA and FOA towards IT outsourcing. In terms of COA, it means the joint participation of both IT departments and business units towards IT outsourcing appropriation, and the two parties reach a consensus on this issue. In other words, COA requires the mutual cooperation between IT managers and business executives towards outsourcing planning and business planning, which is important to improve the business impacts of IT. For example, regular meetings with the IT managers and business managers would strengthen an in-depth understanding regarding the linkage between outsourcing objectives and business goals needed to be supported by IT, and more easily reach an agreement regarding how firms achieve IT alignment through IT outsourcing. In terms of FOA, our empirical results indicated that the structural faithfulness towards IT outsourcing plays a key role in IT alignment. Therefore, to attain IT alignment by using the core technical competencies and advantage resources of suppliers, firms are supposed to appropriate IT outsourcing in a style alignment with the general objectives and procedures intended to be provided by the supplier.

Third, our results also reveal that the moderating effects of SaaS, PaaS, and IaaS on the linkage between outsourcing appropriation and alignment vary significantly. Therefore, to achieve IT alignment, the levels of the COA and FOA should vary depending on the adoption of different cloud computing models. Specifically, our results suggest that when PaaS or IaaS are adopted, the FOA may have a more important impact on attaining IT alignment compared with when they are not adopted. Therefore, when PaaS or IaaS are adopted, it is more critical for outsourcing users to understand the technical characteristics and service

philosophy of the cloud services to ensure a higher level of FOA to a great extent. In contrast, results also indicate that when SaaS or PaaS are adopted, the role of the COA in IT alignment seems to be weakened as compared to when they are not adopted. Thus, when cloud computing is adopted at the application level in terms of SaaS or PaaS, the COA should not be heavily relied upon to promote IT alignment.

5.3. Limitations and future research

Our study has the following limitations that might constrain its generalizability, and provides several topics for future research. First, given that IT alignment is a time-consuming and dynamic process, therefore using a cross-sectional data as our samples to investigate the formation of IT alignment may limit the generalizability of our results. Therefore, future research might adopt a case study or longitudinal design that can reflect the time effect, aiming to reveal the dynamic process of IT alignment through outsourcing appropriation. Second, the alternative antecedents of outsourcing appropriation advocated by AST have not been fully considered in this study, e.g., task characteristics. Meanwhile, we merely focus on the shared IT affordance for knowledge acquisition from the technical perspective. Therefore, future researchers can extend our findings by focusing on other types of antecedents of outsourcing appropriation, e.g., task characteristics or other technology factors (such as sophistication and comprehensiveness). Third, our study employs a unilateral view from outsourcing users to collect data aiming to solve how organizations appropriate IT outsourcing for IT alignment. However, outsourcing suppliers, acted as the direct provider of services, can be seen as another key participant in outsourcing appropriation. Therefore, future research can adopt a bilateral perspective from outsourcing users or suppliers to extend our findings. Fourth, future research could conduct a larger sample survey to further validate our findings. In addition, future research in other contexts, e.g., other countries, could also supplement the findings here.

6. Conclusions

Outsourcing appropriation, covering the two important aspects of COA and FOA, provides a promising lens to answer how to appropriate IT outsourcing for IT alignment. Our study investigates the antecedents of outsourcing appropriation, their impacts on IT alignment, and boundary conditions of the impacts of outsourcing appropriation on IT alignment. Our results demonstrate the importance of shared IT affordance for knowledge acquisition, boundary spanning, and

boundary buffering on outsourcing appropriation apart from the non-significance results of the relationship between boundary buffering and COA. In addition, our findings indicate that COA and FOA contribute to promoting IT alignment respectively, and further advance the literature on the antecedents of suitable outsourcing appropriation. Further, to reveal the boundary conditions of the impacts of outsourcing appropriation on IT alignment, we also consider two environmental factors (degree of outsourcing and cloud computing adoption) as moderators. Our results indicate that the adoption of cloud computing sets the boundary conditions of the linkage between outsourcing appropriation and IT alignment, whereas the degree of outsourcing does not. Our results demonstrate that the moderating effects of different dimensions of cloud computing adoption (including SaaS, PaaS, and IaaS) on the linkage between outsourcing appropriation and IT alignment vary significantly. Notably, the relatively importance of COA and FOA on IT alignment may vary based on different cloud computing adoption mode.

CRedit authorship contribution statement

Tao Wang: Conceptualization, Writing – original draft, Formal analysis, Methodology, Writing – review & editing. **Chun-Ping Deng:** Resources, Conceptualization, Methodology, Investigation, Writing – review & editing, Funding acquisition. **Thompson S.H. Teo:** Writing – review & editing, Conceptualization, Resources, Formal analysis, Methodology.

Declaration of competing interest

None.

Data availability

Data will be made available on request.

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Appendix A

Table A1
Key studies on IT outsourcing and IT alignment.

Authors	Method	Key findings
Ghosh and Scott (2009)	Construction of conceptual model.	The knowledge gaps between outsourcing users and providers may hinder IT alignment.
Valorinta (2011)	Using data obtained from the IT managers and other executives of 500 enterprises	IT outsourcing can improve IT alignment by decreasing IT governance costs. That is, IT outsourcing enables IT function, especially for IT department, to devote more energy to alignment between IT planning and business objectives needed to be supported by IT.
Schmidt and Uydess (2012)	Literature review and construction of conceptual model.	IT outsourcing probably impede IT alignment, in particular, when there exist relationship conflicts between outsourcing providers and users.
Silvius et al. (2013)	A qualitative study with four IT outsourcing cases.	IT outsourcing may impose positive and negative effects on IT business alignment through literature review and case analysis.
Arvanitis et al. (2017)	Using a survey data collected from 556 European firms.	Cloud computing as an emerging outsourcing mode, characterized by pay per use, scalability, and configurability, may enable the higher IT alignment than traditional IT outsourcing.
Park et al. (2017)	A survey-based empirical approach and using data from 213 enterprises that engaged in IT outsourcing businesses.	The integration of internal IT governance (e.g., IT infrastructure or talent) and external IT governance (e.g., cloud computing or IT outsourcing) leads to the best IT alignment.
Lioliou et al. (2019)	Case study within the financial services sector.	The opportunistic behavior of outsourcing providers may impose a negative impact on IT alignment.

(continued on next page)

Table A1 (continued)

Authors	Method	Key findings
Wallenburg et al. (2019)	A survey-based empirical approach and using data from the logistics service industry.	In outsourcing, the interaction between outsourcing suppliers and clients plays a crucial role in promoting IT alignment.
Guan et al. (2020)	Using data from surveys of 240 crowdsources in China.	The relational governance between outsourcing users and providers is a key element of determining the degree of alignment of IT services with business needed to be supported by IT.
Saldanha et al. (2020)	An empirical approach and data obtained from above 120 firms in India.	Some capabilities related to outsourcing management, such as IT planning, delivery and change, may affect IT alignment.
Deng et al. (2021)	Using a survey-based data from the manufacturing and service in China.	IT alignment as a mediating mechanism by which IT outsourcing appropriation impact agility, and thereby may be positively affected by appropriation of IT outsourcing.
Kalubanga and Namagembe (2022)	Using data collected from department executives in 103 manufacturing enterprises.	Whether IT outsourcing promotes or hinders IT alignment is heavily dependent on the relationship quality (e.g., trust and commitment) between outsourcing users and providers.

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